Daniel Winfield

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PROFESSIONAL SUMMARY

M.S. candidate in Mechanical Engineering and Materials Science at Duke University with hands-on experience in experimental research, robotics and embedded systems, computer vision, mechanical design, and materials characterization. Proficient in Python-based engineering analysis, CAD, experimental testing, and multidisciplinary system development.

EDUCATION

Duke University Durham, NC
Master of Science in Mechanical Engineering and Materials Science **Expected December 2026**

- Centennial Graduate Professional Master's MSI Fellowship recipient — full-tuition and stipend support

Tuskegee University Tuskegee, AL
Bachelor of Science in Mechanical Engineering **May 2025**

- Minor in Materials Science & Engineering
- GPA: **3.902/4.0** | **Summa Cum Laude** | **Distinguished Presidential Scholar** | **Dean's List**

ENGINEERING & TECHNICAL PROJECTS

Continuum Manipulator – Lifting Dynamics Analysis **Fall 2026 – Present**
Duke University | Robotics Course Project *Durham, NC*

- Collaborating on the development of a single-section, three-tendon continuum manipulator for studying load-dependent lifting behavior.
- Developing the modeling and experimental approach to evaluate how payload, arm configuration, and lifting speed affect body deformation, positioning accuracy, stability, and usable lifting capacity.

Retractable Y-Zipper Blade Concept **Aug. 2026 – Present**
Personal Engineering Project

- Exploring a compact retractable blade concept using an existing three-strip Y-Zipper flexible-to-rigid mechanism, adapting publicly available geometry and fabrication files as the basis for a personal prototype.
- 3D printed and manually evaluated the original Y-Zipper strips and converger/slider mechanism, testing zipping/unzipping behavior and qualitatively assessing assembled-beam stiffness.
- Investigating the existing Rhino/Grasshopper design workflow and debugging the provided parametric files in preparation for modifying strip geometry for the retractable-blade application.
- Performed preliminary packaging estimates for spool diameter, usable blade length versus handle length, and strip-storage volume to evaluate feasibility of compact retraction.
- Developing an actuation concept using motors and gearing, with initial component exploration based on available embedded-system hardware and plans for dedicated motor/control hardware once the mechanical geometry is established.

PEBBLE – Intelligent Study Companion **Spring 2026**
Duke University | Individual Embedded Systems / AI Course Project *Durham, NC*
Private GitHub Repository

- Developed a Raspberry Pi-based intelligent desk companion integrating sensing, machine learning, computer vision, voice interaction, and real-time multimodal feedback to support focused study sessions.
- Integrated an MPU6050 IMU, ultrasonic sensor, DHT11 temperature sensor, camera, microphone, LCD, buttons, LEDs, buzzer, speaker, and relay-controlled fan.
- Trained a scikit-learn Decision Tree classifier on accelerometer and gyroscope features to classify user motion as focused or distracted, achieving approximately **92% training accuracy and 85% test accuracy**.
- Implemented YOLOv8-based phone detection and combined vision with ultrasonic sensing to identify user presence and distraction.

- Developed real-time control logic using a central state machine, motion smoothing, majority voting, distraction thresholds, presence overrides, and temperature-control hysteresis.
- Integrated a voice-assistant pipeline using **Vosk speech-to-text, GPT API integration, and Piper text-to-speech**, including conversational follow-up.
- Developed multiple operating modes, including a focus timer, voice assistant, reaction game, Simon Says, and music mode, and organized the Python system into modular sensing, AI, control, UI, game, and hardware components.

SurviveIt – Interactive Disaster-Survival Game **Spring 2026**
Duke University | Embedded Systems Group Project *Durham, NC*
Private GitLab Repository

- Collaborated on a Raspberry Pi-based interactive disaster-survival game integrating computer vision, LLM-generated scenarios, physical controls, displays, audio, and electromechanical feedback.
- Led much of the physical hardware integration and assembly, including wiring, enclosure setup, touchscreen and LCD installation, buttons, LEDs, buzzer, ultrasonic sensing, and servo-actuated outcome flags.
- Adapted previously developed Python sensor and voice-interface code for the group system, including ultrasonic sensing and Piper text-to-speech.
- Helped integrate YOLOv8 person detection using a Raspberry Pi camera to detect a nearby user and automatically initiate gameplay.
- Integrated a **5-inch 800×480 DSI touchscreen** and **16×2 I²C LCD** with GPIO-based physical controls for synchronized display and user interaction.
- Supported system-wide integration, testing, and debugging across hardware, vision, voice, sensing, displays, and game logic.
- Contributed to a completed system featuring **seven disaster scenarios**, LLM-generated choices and consequences, narrated feedback, and servo-driven green/red flags indicating survival outcomes.

Autonomous Chess-Playing Robot **Fall 2025**
Duke University | Robotics Course Project *Durham, NC*
Portfolio

- Collaborated on a five-person team to develop an autonomous chess-playing robot integrating computer vision, game logic, ROS 2 motion planning, and robotic manipulation.
- Trained and evaluated a custom **YOLOv8** model for chess-piece detection and classification, achieving approximately **99% detection accuracy** across piece classes.
- Prepared and augmented training data using grayscale conversion, rotation, resizing, and brightness adjustment to improve robustness to real-world lighting and camera conditions.
- Collected and curated physical-board images and video for testing and validation, helping address the domain gap between the original image dataset and real-world operation.
- Supported integration and testing of the vision subsystem within the broader robotic architecture using **ArUco localization, Stockfish/python-chess, MoveIt, ROS 2, and a Kinova Gen3 Lite** robotic arm.
- Contributed to a completed system capable of detecting board state, selecting legal moves, and completing full autonomous games against a human opponent.
- Produced and edited the project's final gameplay demonstration video and supporting media.

ThirdEye – Computer Vision Ptosis Measurement System **Fall 2025**
Duke University | Engineering Project *Durham, NC*
Portfolio

- Co-developed the computer vision and machine-learning subsystem for a mobile tool designed to automate eyelid measurements used in clinical evaluation of ptosis.
- Developed and evaluated a **DeepLabV3 semantic-segmentation model with a ResNet-101 backbone** in Python/PyTorch to identify periocular features including the eyelids, iris, sclera, brows, and pupil.
- Processed and analyzed annotated facial-image data from the **Chicago Facial Dataset** and **CelebAMask-HQ** to support automated calculation of **MRD1, MRD2, and palpebral fissure height**.
- Evaluated segmentation performance using Dice-based metrics and compared the DeepLabV3 approach with a teammate-developed **U-Net/MobileNetV3** architecture.
- Developed and configured a **Raspberry Pi 4B-based prototype phone** with a 5-inch touchscreen, 1080p camera, and Android 15, including display configuration, DPI adjustment, orientation, and real-device application testing.

- Contributed to a final prototype that produced eyelid measurements within approximately **1–2 mm** of ruler-estimated measurements, verified by an ophthalmologist, with results returned in approximately **10–15 seconds** under a stable network connection.

JellyGrip – Biomimetic Robotic End Effector **Fall 2024**
Tuskegee University | Senior Capstone Design Project *Tuskegee, AL*
Portfolio

- Designed a biomimetic robotic end effector inspired by jellyfish tentacles for adaptive manipulation of objects with varying shapes, sizes, and materials in warehouse-automation applications.
- Led the conceptual and detailed design of articulated finger structures with proposed inflatable pads, including finger geometry, bending behavior, mechanical constraints, and integration with the gripper; the inflation system remained under development.
- Developed detailed CAD models and full-system assemblies incorporating articulated fingers, proposed inflatable pads, a PLA structural frame, and a stainless-steel lead screw/brass nut actuation mechanism.
- Performed theoretical gripping-force, pressure, torque, weight, and safety-factor calculations against a 5 lb handling target and safety factor of 2; physical load capacity was not tested.
- Conducted an **FMEA** addressing tentacle rupture, lead-screw binding, structural deformation, and pneumatic leakage, and developed corresponding mitigation and testing strategies.
- Developed a conceptual manufacturing and prototyping plan using FDM-printed PLA components, flexible vinyl tentacles, off-the-shelf hardware, and modular assembly.
- Completed the conceptual design and theoretical-analysis phase, producing a mechanical CAD assembly, cost analysis, manufacturing strategy, and recommendations for future physical prototyping.

Advanced Materials Laboratory – Polymer & Composite Characterization **Spring 2024**
Tuskegee University | Advanced Materials Laboratory Course *Tuskegee, AL*
Portfolio

- Fabricated **CF-PETG** test specimens using FDM additive manufacturing, creating geometry in Fusion 360 and preparing builds in PrusaSlicer while varying infill density from **20–100%**.
- Performed tensile and three-point flexural testing on 3D-printed CF-PETG, analyzing stress-strain behavior, strength, and stiffness.
- Conducted **Dynamic Mechanical Analysis (DMA)** on 3D-printed PLA to characterize storage modulus, loss modulus, and temperature-dependent viscoelastic behavior, identifying a glass-transition temperature of approximately **74.8°C**.
- Used **Thermogravimetric Analysis (TGA)** to evaluate PLA thermal stability and degradation behavior, identifying decomposition over approximately **350–415°C**.
- Applied **X-ray diffraction (XRD)** and Bragg's Law to investigate crystallinity and interatomic spacing in a carbon-based sample.
- Characterized PLA using **optical microscopy and Differential Scanning Calorimetry (DSC)** to examine morphology, orientation, surface characteristics, and thermal transitions, identifying crystallization and melting temperatures near **105°C and 170°C**, respectively.

RESEARCH EXPERIENCE

Graduate Researcher – Thermal Diffusion in Architected Digital Composites **2026 – Present**
Ni Lab, Duke University *Durham, NC*

- Investigating thermal diffusion and transient heat-transfer behavior in architected digital composites, with emphasis on how material properties and spatial architecture influence surface-temperature evolution and heat propagation.
- Developing an experimental approach for characterizing the electrical-to-thermal response of liquid-metal digital-composite elements and measuring transient surface-temperature behavior.
- Previously supported soft robotic fish-tail research through experimental and computer-vision-based motion and flow characterization, including **stereo tracking and Particle Image Velocimetry (PIV)**.
- Developed and used Python-based video-analysis workflows for tracking motion, processing experimental data, and analyzing time-dependent and frequency-domain behavior.

Undergraduate Research Intern – Quantum Technologies & Computational Materials Design **Jan. 2024 – Mar. 2025**
Tuskegee University, Department of Physics *Tuskegee, AL*
Portfolio

- Conducted DOE-supported computational-materials research investigating point defects in cubic beryllium sulfide (BeS) as potential solid-state systems for spin-based quantum technologies.
- Applied **Density Functional Theory (DFT), GW, and Bethe–Salpeter Equation (BSE)** methods to pristine and defected cubic BeS to evaluate ground-state electronic structure and excited-state optical behavior.
- Analyzed electronic density of states, band gaps, optical excitation spectra, structural stability, and defect-induced lattice distortions across vacancy and dopant-modified configurations.
- Determined that the three investigated defect configurations exhibited **triplet ground states and dynamical stability**, with intense low-energy optical excitations near **0.5 eV**.
- Contributed to a research hypothesis relating under-coordinated electronegative atoms and electron parity to the formation of triplet ground states in local defects.
- Used **Fortran, computational data analysis, and molecular/crystal visualization** to interpret defect structures and electronic and optical behavior.
- Collaborated with researchers at Tuskegee University and the University of Central Florida and presented the work at the **APS March Meeting 2025 in Anaheim, California**.

Research Assistant – Hybrid Fiber-Reinforced Polymer Composites **Jan. 2023 – May 2023**
Tuskegee University | Materials Science *Tuskegee, AL*

- Investigated the mechanical and viscoelastic behavior of **jute, E-glass, and hybrid jute/glass fiber-reinforced epoxy composites** to evaluate the effects of combining natural and synthetic reinforcement.
- Fabricated composite laminates using **wet hand lay-up and compression molding** with a biobased epoxy matrix and prepared specimens for mechanical and dynamic testing.
- Performed **DMA, tensile, and flexural testing** to compare storage modulus, loss modulus, damping behavior, tensile strength, and flexural strength across the three composite systems.
- Analyzed results showing that the hybrid composite exceeded the jute-only composite in mechanical performance while providing intermediate behavior between the natural-fiber and E-glass systems.
- Evaluated DMA behavior, including storage modulus and tan-delta response, to assess stiffness and energy-dissipation characteristics.
- Coauthored the project report and presented findings on fabrication methods, experimental results, and the potential influence of fiber stacking order on future hybrid-composite performance.

PROFESSIONAL EXPERIENCE

Student Gardener **Jan. 2026 – Present**
Sarah P. Duke Gardens, Duke University *Durham, NC*

- Support routine garden and grounds maintenance, including planting, weeding, pruning, mulching, irrigation, material handling, and site setup and cleanup.
- Use a variety of gardening hand tools to maintain planted areas and assist with seasonal landscape work throughout the Gardens.
- Work both independently and with staff crews to complete assigned maintenance tasks and support day-to-day garden operations.

Project Execution Co-op **Jun. 2023 – Aug. 2023**
Honda Development & Manufacturing of America *Raymond, OH*

- Supported project-execution and quality-related activities within the Quality Department of Honda's R&D facility.
- Migrated associate training information from an existing spreadsheet-based tracking process into **Power BI**, developing a centralized dashboard for organizing and monitoring training records across departments.
- Structured and visualized training data to improve accessibility and provide clearer visibility into associate training status and completion.
- Supported plant-floor Quality personnel on multiple occasions, including production-related activities involving vehicle hood components.
- Participated in tours of Honda manufacturing, R&D, and supporting operations, gaining exposure to automotive production processes, quality workflows, and cross-functional engineering activities.

- Completed Honda internal training covering manufacturing operations, quality practices, workplace procedures, and Lean/Kaizen concepts.

Human Resources / Student Records Intern **Jun. 2022 – Jul. 2022**
Wallace Community College Selma *Selma, AL*

- Supported daily operations in the Student Admissions and Records department.
- Converted physical student records into organized digital files to support recordkeeping and document accessibility.
- Assisted students with requests for transcripts, academic records, and degree-related documentation and routed paperwork to the appropriate departments.

TEACHING & LEADERSHIP EXPERIENCE

Teaching Assistant – Measurement & Analysis Laboratory **Jan. 2025 – May 2025**
Tuskegee University *Tuskegee, AL*

- Set up, operated, and troubleshot laboratory experiments and instrumentation used for undergraduate measurement and analysis instruction, including thermocouples, strain/load measurement, National Instruments data-acquisition hardware, and fluid-mechanics experiments.
- Assisted students with experimental procedures, sensor calibration, measurement techniques, data acquisition, uncertainty/error analysis, and interpretation of engineering results.
- Led hands-on laboratory demonstrations and provided instruction on proper experimental practices, instrumentation use, and engineering measurement concepts.
- Provided one-on-one and group technical guidance, helping students diagnose experimental issues and connect measured results to underlying engineering principles.

HONORS, AWARDS & PROFESSIONAL DEVELOPMENT

MITdesignX – Design Innovation & Entrepreneurship Workshop **Jan. 2025**
Massachusetts Institute of Technology *Cambridge, MA*

- Selected as one of four Tuskegee University Mechanical Engineering students to participate in a week-long MIT-hosted design innovation and entrepreneurship program.
- Collaborated on the design and physical prototyping of **Booker**, a wooden portable organization case for architects with dedicated storage for a ruler, compass, pens, and other tools.
- Developed the concept through needs analysis, stakeholder and market evaluation, problem-solution fit, value-proposition development, Rhino-based design preparation, and hands-on woodshop fabrication with CNC exposure.
- Incorporated the historical connection between architect **Robert R. Taylor, MIT, and Tuskegee** into the product concept and storytelling, then refined the prototype and delivered a final pitch/showcase.

NSF Louis Stokes Alliance for Minority Participation (LSAMP) Scholar **Aug. 2021 – May 2025**
TMCF / Novartis Career Readiness & Mentoring Program Scholar **Aug. 2022 – May 2025**
Novartis US Foundation & Thurgood Marshall College Fund

- Selected for a multi-year scholarship and professional-development program supporting HBCU students through career-readiness, mentorship, and industry-engagement resources.

IBM Accelerate – Hardware Track **Summer 2023**
IBM

- Completed an **8-week technical learning and mentorship program** focused on hardware careers, with exposure to processor and hardware development, product engineering, firmware development, and supply-chain engineering.
- Earned the **IBM Accelerate – Hardware digital credential** upon program completion.

TCS PacePort Innovation Center – Selected Student Participant **Oct. 2022**
Tata Consultancy Services *New York, NY*

- Selected for a two-day industry immersion featuring discussions on emerging technologies, innovation, leadership, and the future of work with TCS professionals and senior leadership.

GE Appliances & Tuskegee Alumni Career Readiness and Leadership Development Program **Feb. – Apr. 2022**

- Completed a multi-session professional-development program covering professional performance and image, networking, employer follow-up, presentation skills, and career readiness.

TECHNICAL SKILLS

- **Programming, Data & Software:** Python, C++, Fortran, MATLAB; NumPy, Pandas, Matplotlib; object-oriented and modular programming; automated data processing and visualization; Microsoft Excel, Power BI
- **Software Development & Computing:** Git, GitHub, GitLab, Linux, Bash/command line, SSH, Conda, pip, Python virtual environments, Docker; REST API and JSON integration; basic networking and remote-system configuration
- **Machine Learning & Computer Vision:** PyTorch, scikit-learn, OpenCV, Ultralytics YOLOv8, DeepLabV3, ResNet-101; semantic segmentation, object detection, image preprocessing and augmentation, feature extraction, classification, video tracking
- **Computer Vision & Optical Measurement:** Camera calibration, stereo vision, ArUco/ChArUco markers, homography and perspective transformation, image-based measurement, triangulation, coordinate mapping, motion tracking, Particle Image Velocimetry (PIV)
- **Robotics & Embedded Systems:** ROS 2, MoveIt, RViz, Raspberry Pi, Arduino; forward/inverse kinematics, coordinate transformations, Jacobians, robot dynamics, state-machine design; GPIO, I²C, UART/serial, PWM, sensor integration, motor and relay control
- **Electronics & Instrumentation:** Breadboarding, circuit prototyping, schematic interpretation, soldering, digital multimeters, oscilloscopes, bench power supplies, National Instruments DAQ hardware, thermocouples, strain gauges, load cells, sensor calibration; DC, stepper, and servo motors; H-bridges, motor drivers, ESCs; electrical and hardware/software troubleshooting
- **CAD, Mechanical Design & Prototyping:** Autodesk Fusion 360, Siemens NX, Rhino, Grasshopper; engineering drawings, tolerance analysis, fits and clearances, material selection, requirements definition, trade studies, FMEA, DFM/DFA, rapid prototyping, iterative design, mechanical assembly
- **Additive Manufacturing & Fabrication:** FDM 3D printing using Creality/Ender and Prusa systems; Cura, PrusaSlicer, Bambu Studio; print setup, slicing, orientation/support optimization, troubleshooting; machine-shop and fabrication training including manual mill, lathe, drill press, bandsaw, sanding equipment, CNC equipment, waterjet, laser cutter, and hand/power tools
- **Materials Processing & Characterization:** Polymer and fiber-reinforced composites, wet hand lay-up, compression molding, specimen preparation; tensile and three-point flexural testing, DMA, TGA, DSC, XRD, optical microscopy; ASTM-based mechanical and materials testing
- **Computational Materials:** Density Functional Theory (DFT), GW, Bethe–Salpeter Equation (BSE) methods; electronic-structure, defect, density-of-states, and optical-property analysis; crystalline and molecular visualization
- **Experimental Methods & Data Analysis:** Experimental setup and calibration, measurement uncertainty/error analysis, repeatability testing, calibration curves, regression and curve fitting, descriptive statistics, correlation, confidence intervals, hypothesis testing, controlled test-matrix development, time-series analysis, FFT/frequency-domain analysis, vibration analysis, automated experimental-data processing
- **Thermal & Fluid Systems:** Heat-transfer fundamentals including conduction, convection, radiation, and transient thermal behavior; building heat-transfer and thermal-load simulation; thermal imaging/IR-camera experience; fluid-mechanics experimentation, flow measurement, and pressure measurement
- **Engineering & Project Development:** Root-cause analysis, Lean/Kaizen concepts, Gantt charts, project scheduling, task coordination, milestone tracking, technical literature review, technical documentation, engineering reports, research presentations, technical instruction, mentoring, and team leadership
- **Additional Exposure:** SolidWorks, finite element analysis (FEA), PID and closed-loop control, URDF/Xacro, CNC machining, introductory GD&T concepts, and SPI communication

PROFESSIONAL AFFILIATIONS

- National Society of Black Engineers (NSBE)
- Society of Manufacturing Engineers (SME)
- American Physical Society (APS)